(AUDREY-LORAINE) CHLOÉ TANAKA

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Harvard Kennedy School, 79 John F. Kennedy Street, Cambridge, MA 02138

Education

Harvard University
Ph.D. Public Policy, Economics Track, Expected Completion May 2027
Committee: Gordon Hanson (Chair), Joseph Aldy, Joshua Goodman

University of Cambridge
M.Phil. Economics, 2016

University of Alaska Anchorage
B.A. Economics, Minor Mathematics, *summa cum laude,* 2013

Georgetown University
B.A. Economics, Minor Mathematics, *First Honors,* Transferred December 2010

Fields

Primary Field: Environmental and Energy Economics
Secondary Fields: Labor Economics, Education, Public Economics

Teaching

Principles of Economics – Macroeconomics (ECON 10b), Harvard University, 2024
Teaching Fellow for Professor David Laibson and Professor Jason Furman

Principles of Economics – Microeconomics (ECON 10a), Harvard University, 2023
Teaching Fellow for Professor David Laibson and Professor Jason Furman

Relevant Experience

Doctoral Researcher, Reimagining the Economy, Gordon Hanson, 2022-Present

Credit Associate, Bridgepoint Capital, 2019-2021

Investment Banking Analyst, Barclays Investment Bank, 2017-2019

Research Professional, UAA, Institute of Social and Economic Research, 2013-2015

Research Assistant, UAA, Experimental Economics Lab, 2011-2013

Research In Progress

Powering the Workforce: The Role of Community Colleges in Energy Transformations

The pace of energy transformations depends not only on technology and capital, but also on the supply of skilled labor. Meeting rapidly growing electricity demand and decarbonization goals will require swift expansion of this workforce, yet whether local training institutions are equipped to deliver it remains an open question. Using the U.S. shale boom as a natural experiment, I estimate the causal effect of energy-induced changes in local labor demand on vocational programs and graduations at community colleges. Exploiting staggered boom timing across plays, I use stacked difference-in-differences and event-study designs. I find that colleges inside booming plays expand energy programs by 26% and energy completions by 111% relative to baseline, driven almost entirely by newly created programs. The response is slow but persistent, emerging several years after the boom and continuing for over a decade. Consistent with rising opportunity costs of enrollment, non-energy fields stagnate, with 10% fewer programs and 25% fewer completions than expected absent the boom. Extending the framework to the clean energy transition, I find that a one-gigawatt increase in local solar investment raises solar programs by 28% and completions by 37%. Community colleges can help train the energy workforce, but their adjustment is gradual, uneven, and costly to other fields.

Blowin' in the Wind: The Impact of Climate Disasters on College Majors and Innovation (with Avinash Moorthy)

Could extreme weather influence education and career choices? This project investigates how foundational experiences with extreme weather shape human capital formation and innovative activity. We study the causal impact of exposure to extreme weather on college major choices, as well as on the direction, quality, and intensity of inventors' output, examining the extent to which shifts in educational pathways serve as a mediating channel. By linking large-scale data on natural disasters to individual-level patent and education records, our research aims to provide new evidence on how personal experiences shape career trajectories and the evolution of technical progress.

The Effect of Coal Power Plant Retirements on Labor Market Outcomes (with Gordon Hanson)

Publications

"Short-run impacts of a severance tax change: Evidence from Alaska."
Energy Policy, Volume 107, 2017 (with Matthew Reimer and Mouhcine Guettabi)

Policy Reports

"The economic geography of salmon: A conceptual framework and preliminary characterization of the spatial distribution of economic values associated with salmon in the Mat-Su Basin."
Alaska Department of Fish and Game, 2015 (with David Albert, David Holen, Bronwyn Jones, and Tobias Schwörer)

"Alaskan cod market analysis."
Bristol Bay Economic Development Corporation, 2013 (with Matthew Reimer)

Fellowships, Honors, & Awards

Harvard University Graduate Prize Fellowship, 2021

Marshall Scholarship Finalist, 2014

UAA Garth N. Jones Undergraduate Award for Excellence in Writing, 2014

UAA Award for Academic Excellence, 2013

UAA Bradford Tuck Senior in Economics Scholarship, 2011-2012

Seminars

2026: Harvard Environmental Economics Workshop; Harvard Labor and Public Economics Workshop; Harvard Economics and Social Policy Workshop

2025: Harvard Economics and Social Policy Workshop

Invited Conferences & Workshops

CERAWeek NextGen Cohort Poster Presentation, 2026

Carnegie Mellon University Engineering and Public Policy, Doctoral Student Participatory Workshop on Climate and Energy Decision Making, 2026

"Aces high or low? The impact of a severance tax change on Alaskan oil activity."
Western Regional Science Association (WRSA), 2014 *(presented by coauthor)*

"Seasonal variation in risk preferences."
Economic Science Association (ESA), 2012 *(presented by coauthor)*

Affiliations

Doctoral Student Fellow, Harvard Environmental Economics Program

Professional Service

Peer Mentor, Kennedy PhD Student Association (KPSA), 2024-2025
Social Chair, Kennedy PhD Student Association (KPSA), 2022-2023

Software Skills

Stata, MATLAB, ArcGIS, Python, R, LaTeX